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In [4]: %pip install numpy pandas matplotlib seaborn
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Collecting pandas
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Collecting fonttools>=4.28.2 (from matplotlib)
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[illegible]

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```
Installing collected packages: pyparsing, pillow, numpy, kiwisolver, fonttools, cycler, pandas, contourpy, matplotlib, seaborn
```

----- 0/10 [pyparsing]

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[illegible]

Successfully installed contourpy-1.4.0 cycycler-0.12.1 fonttools-4.65.0 kiwisolver-1.5.1 matplotlib-3.11.2 numpy-2.5.3 pandas-3.0.5 pillow-12.3.0 pyparsing-3.3.2 seaborn-0.13.2

Note: you may need to restart the kernel to use updated packages.

WARNING: The scripts f2py.exe and numpy-config.exe are installed in 'C:\Users\vimal\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The scripts fonttools.exe, pyftmerge.exe, pyftsubset.exe and ttx.exe are installed in 'C:\Users\vimal\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

```
In [6]: import sqlite3
import pandas as pd

# 1. Connect to the database
conn = sqlite3.connect('customer_churn.db')

# 2. Fetch table names (fixed unclosed quote)
sql_query = "SELECT name FROM sqlite_master WHERE type = 'table';"
tables = pd.read_sql(sql_query, conn)

# 3. Store DataFrames in a dictionary (recommended best practice)
dfs = {}
for table_name in tables['name']:
    # Fixed missing space after 'FROM'
    dfs[table_name] = pd.read_sql(f"SELECT * FROM {table_name}", conn)
    print(f"Loaded DataFrame for: {table_name}")

# 4. Close connection AFTER the loop finishes
conn.close()

# Access individual DataFrames via dict keys, e.g., dfs['customers']
```

Loaded DataFrame for: db\_customer

Loaded DataFrame for: db\_subscription

Loaded DataFrame for: db\_support

```
In [7]: # Print table names and column names
conn = sqlite3.connect('customer_churn.db')

for table_name in tables['name']:
    print(f"\nTable Name: {table_name}")
    # Get column information
    columns_query = f"PRAGMA table_info({table_name});"
    columns = pd.read_sql(columns_query, conn)
    print("Columns:")
    print(columns['name'].tolist())

# Close connection
conn.close()
```

Table Name: db\_customer

Columns:

['customerid', 'name', 'country', 'state', 'gender', 'dob', 'interests', 'pincode']

Table Name: db\_subscription

Columns:

['customerid', 'subscription\_start\_date', 'subscription\_type', 'renewal\_date', 'plan\_type', 'contract\_type', 'cancellation\_date', 'cancellation\_reason', 'monthly\_charges', 'cltv', 'churn\_score']

Table Name: db\_support

Columns:

['customerid', 'complaint\_date', 'escalations', 'csat\_score', 'col\_1', 'comment']

```
In [9]: import sqlite3
import pandas as pd

conn = sqlite3.connect('customer_churn.db')

# Fetch table names
tables = pd.read_sql("SELECT name FROM sqlite_master WHERE type='table';", conn)

# Print exact names present in your database
print("Tables found:", tables['name'].tolist())

# Store in a dictionary
dfs = {}
for table_name in tables['name']:
    dfs[table_name] = pd.read_sql(f"SELECT * FROM {table_name}", conn)

conn.close()

# Preview the desired table using its exact key name from the printout above
# Example: dfs['customer'].head()
```

Tables found: ['db\_customer', 'db\_subscription', 'db\_support']

```
In [11]: dfs['db_customer'].head()
```

```
Out[11]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>customerid</th> <th>name</th> <th>country</th> <th>state</th>
<th>gender</th> <th>dob</th> <th>interests</th> <th>pincode</th> </thead>
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|            |        |       |             |        |                     |        |      |
|------------|--------|-------|-------------|--------|---------------------|--------|------|
| 0002-ORFBO | keshav | India | Maharashtra | Male   | 1982-04-12 00:00:00 | travel | None |
| 0003-MKNFE | raghav | India | Karnataka   | Male   | 1995-11-23 00:00:00 | NaN    | None |
| 0004-TLHLJ | lalita | India | Delhi       | Female | 1978-02-15 00:00:00 | movie  | None |
| 0011-IGKFF | mohan  | India | Nagaland    | Male   | 2001-08-30 00:00:00 | NaN    | None |
| 0013-EXCHZ | mira   | India | Delhi       | Female | 1990-05-05 00:00:00 | drama  | None |

```
In [12]: #a. rename col_name
#b. drop coloumns_interest and oincode
```

```
#c. change data type_dob
#d. data standardization_gender
#e. fix mixing values_country
```

```
In [15]: #a. rename col_name
dfs['db_customer'].rename(columns={'name': 'customer_name'}, inplace=True)
```

```
In [17]: dfs['db_customer'].head()
```

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Out[17]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
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<th>state</th> <th>gender</th> <th>dob</th> <th>interests</th>
<th>pincode</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th>
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| 0002-ORFBO | keshav | India | Maharashtra | Male   | 1982-04-12 00:00:00 | travel | None |
| 0003-MKNFE | raghav | India | Karnataka   | Male   | 1995-11-23 00:00:00 | NaN    | None |
| 0004-TLHLJ | lalita | India | Delhi       | Female | 1978-02-15 00:00:00 | movie  | None |
| 0011-IGKFF | mohan  | India | Nagaland    | Male   | 2001-08-30 00:00:00 | NaN    | None |
| 0013-EXCHZ | mira   | India | Delhi       | Female | 1990-05-05 00:00:00 | drama  | None |

```
In [20]: dfs['db_customer'] = dfs['db_customer'].drop(columns=['interest', 'pincode'], er
```

```
In [21]: dfs['db_customer'].head()
```

```
Out[21]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
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<th>state</th> <th>gender</th> <th>dob</th> <th>interests</th> </thead>
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| 0002-ORFBO | keshav | India | Maharashtra | Male   | 1982-04-12 00:00:00 | travel |
| 0003-MKNFE | raghav | India | Karnataka   | Male   | 1995-11-23 00:00:00 | NaN    |
| 0004-TLHLJ | lalita | India | Delhi       | Female | 1978-02-15 00:00:00 | movie  |
| 0011-IGKFF | mohan  | India | Nagaland    | Male   | 2001-08-30 00:00:00 | NaN    |
| 0013-EXCHZ | mira   | India | Delhi       | Female | 1990-05-05 00:00:00 | drama  |

```
In [22]: dfs['db_customer'] = dfs['db_customer'].drop(columns=['interest', 'pincode'], er
```

```
In [23]: dfs['db_customer'].head()
```

Out[23]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer\_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> <th>interests</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> </tbody></tbody>

|            |        |       |             |        |                     |        |
|------------|--------|-------|-------------|--------|---------------------|--------|
| 0002-ORFBO | keshav | India | Maharashtra | Male   | 1982-04-12 00:00:00 | travel |
| 0003-MKNFE | raghav | India | Karnataka   | Male   | 1995-11-23 00:00:00 | NaN    |
| 0004-TLHLJ | lalita | India | Delhi       | Female | 1978-02-15 00:00:00 | movie  |
| 0011-IGKFF | mohan  | India | Nagaland    | Male   | 2001-08-30 00:00:00 | NaN    |
| 0013-EXCHZ | mira   | India | Delhi       | Female | 1990-05-05 00:00:00 | drama  |

In [24]: `dfs['db_customer'] = dfs['db_customer'].drop(columns=['interests', 'pincode'], e`

In [25]: `dfs['db_customer'].head()`

Out[25]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer\_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> </tbody></tbody>

|            |        |       |             |        |                     |
|------------|--------|-------|-------------|--------|---------------------|
| 0002-ORFBO | keshav | India | Maharashtra | Male   | 1982-04-12 00:00:00 |
| 0003-MKNFE | raghav | India | Karnataka   | Male   | 1995-11-23 00:00:00 |
| 0004-TLHLJ | lalita | India | Delhi       | Female | 1978-02-15 00:00:00 |
| 0011-IGKFF | mohan  | India | Nagaland    | Male   | 2001-08-30 00:00:00 |
| 0013-EXCHZ | mira   | India | Delhi       | Female | 1990-05-05 00:00:00 |

In [27]: `dfs['db_customer']['dob'] = pd.to_datetime(dfs['db_customer']['dob'], errors='co`

In [28]: `dfs['db_customer'].head()`

Out[28]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer\_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> </tbody></tbody>

|            |        |       |             |        |            |
|------------|--------|-------|-------------|--------|------------|
| 0002-ORFBO | keshav | India | Maharashtra | Male   | 1982-04-12 |
| 0003-MKNFE | raghav | India | Karnataka   | Male   | 1995-11-23 |
| 0004-TLHLJ | lalita | India | Delhi       | Female | 1978-02-15 |
| 0011-IGKFF | mohan  | India | Nagaland    | Male   | 2001-08-30 |
| 0013-EXCHZ | mira   | India | Delhi       | Female | 1990-05-05 |

In [30]: `dfs['db_customer'].info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   customerid      21 non-null    str
1   customer_name   21 non-null    str
2   country         18 non-null    str
3   state           21 non-null    str
4   gender          21 non-null    str
5   dob             21 non-null    datetime64[us]
dtypes: datetime64[us](1), str(5)
memory usage: 1.1 KB
```

In [32]: `dfs['db_customer']['gender'].unique()`

Out[32]: <StringArray>  
['Male', 'Female', 'Women', 'Men']  
Length: 4, dtype: str

In [35]: `mapping = {`  
 `'Men': 'Male', 'M': 'Male',`  
 `'Women': 'Female', 'F': 'Female'`  
`}`  
`dfs['db_customer']['gender'] = dfs['db_customer']['gender'].replace(mapping)`

In [36]: `dfs['db_customer']['gender'].unique()`

Out[36]: <StringArray>  
['Male', 'Female']  
Length: 2, dtype: str

In [38]: `dfs['db_customer'][dfs['db_customer']['country'].isna()]`

Out[38]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>  
<th></th> <th>customerid</th> <th>customer\_name</th> <th>country</th>  
<th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>5</th>  
<th>8</th> <th>12</th> </tbody> <tbody></tbody>  
0013-MHZWF durga NaN Delhi Female 1988-12-10  
0015-UOCOJ maya NaN Kathmandu Female 1985-07-07  
0018-NYROU chitra NaN Telangana Female 2004-12-01

In [40]: `dfs['db_customer'][dfs['db_customer']['country'].isna()]`

Out[40]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer\_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>5</th> <th>8</th> <th>12</th> </tbody> <tbody></tbody>

|            |       |     |       |        |            |
|------------|-------|-----|-------|--------|------------|
| 0013-MHZWF | durga | NaN | Delhi | Female | 1988-12-10 |
|------------|-------|-----|-------|--------|------------|

|            |      |     |           |        |            |
|------------|------|-----|-----------|--------|------------|
| 0015-UOCOJ | maya | NaN | Kathmandu | Female | 1985-07-07 |
|------------|------|-----|-----------|--------|------------|

|            |        |     |           |        |            |
|------------|--------|-----|-----------|--------|------------|
| 0018-NYROU | chitra | NaN | Telangana | Female | 2004-12-01 |
|------------|--------|-----|-----------|--------|------------|

In [41]: `dfs['db_customer']['country'] = dfs['db_customer']['country'].fillna('Nahi Pata')`

In [42]: `dfs['db_customer'][dfs['db_customer']['country'].isna()]`

Out[42]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer\_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> </tbody> <tbody></tbody>

In [46]: `dfs['db_customer']['country'] = dfs['db_customer']['country'].fillna('Nahi Pata')`

In [47]: `dfs['db_customer'][dfs['db_customer']['country'] == 'Nahi Pata']`

Out[47]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer\_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>5</th> <th>8</th> <th>12</th> </tbody> <tbody></tbody>

|            |       |           |       |        |            |
|------------|-------|-----------|-------|--------|------------|
| 0013-MHZWF | durga | Nahi Pata | Delhi | Female | 1988-12-10 |
|------------|-------|-----------|-------|--------|------------|

|            |      |           |           |        |            |
|------------|------|-----------|-----------|--------|------------|
| 0015-UOCOJ | maya | Nahi Pata | Kathmandu | Female | 1985-07-07 |
|------------|------|-----------|-----------|--------|------------|

|            |        |           |           |        |            |
|------------|--------|-----------|-----------|--------|------------|
| 0018-NYROU | chitra | Nahi Pata | Telangana | Female | 2004-12-01 |
|------------|--------|-----------|-----------|--------|------------|

In [48]: `dfs['db_customer']['country'] = dfs['db_customer']['country'].fillna('None')`

In [49]: `dfs['db_customer'][dfs['db_customer']['country'] == 'Nahi Pata']`

Out[49]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer\_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>5</th> <th>8</th> <th>12</th> </tbody> <tbody></tbody>

|            |       |           |       |        |            |
|------------|-------|-----------|-------|--------|------------|
| 0013-MHZWF | durga | Nahi Pata | Delhi | Female | 1988-12-10 |
|------------|-------|-----------|-------|--------|------------|

|            |      |           |           |        |            |
|------------|------|-----------|-----------|--------|------------|
| 0015-UOCOJ | maya | Nahi Pata | Kathmandu | Female | 1985-07-07 |
|------------|------|-----------|-----------|--------|------------|

|            |        |           |           |        |            |
|------------|--------|-----------|-----------|--------|------------|
| 0018-NYROU | chitra | Nahi Pata | Telangana | Female | 2004-12-01 |
|------------|--------|-----------|-----------|--------|------------|

In [50]: `dfs['db_customer'][['country', 'state']]`

Out[50]: `<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>country</th> <th>state</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>`

|           |               |
|-----------|---------------|
| India     | Maharashtra   |
| India     | Karnataka     |
| India     | Delhi         |
| India     | Nagaland      |
| India     | Delhi         |
| Nahi Pata | Delhi         |
| India     | Meghalaya     |
| India     | Rajasthan     |
| Nahi Pata | Kathmandu     |
| Nepal     | Kathmandu     |
| India     | Maharashtra   |
| India     | Karnataka     |
| Nahi Pata | Telangana     |
| India     | Meghalaya     |
| India     | Uttar Pradesh |
| India     | Delhi         |
| India     | Meghalaya     |
| India     | Rajasthan     |
| India     | Telangana     |
| India     | Uttar Pradesh |
| India     | Maharashtra   |

In [52]: `# country and state - unique value pair  
state_country_mapping = dfs['db_customer'].dropna(subset=['country']).set_index(  
dfs['db_customer']['country'] = dfs['db_customer']['country'].fillna(dfs['db_cus`

In [53]: `dfs['db_customer'][dfs['db_customer']['country'].isna()]`

```
Out[53]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>customerid</th> <th>customer_name</th> <th>country</th>
<th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> </tbody>
<tbody></tbody>
```

```
In [54]: dfs['db_customer'][['country', 'state']]
```

```
Out[54]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>country</th> <th>state</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th>
<th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th>
<th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th>
<th>19</th> <th>20</th> </tbody> <tbody></tbody>
```

|           |               |
|-----------|---------------|
| India     | Maharashtra   |
| India     | Karnataka     |
| India     | Delhi         |
| India     | Nagaland      |
| India     | Delhi         |
| Nahi Pata | Delhi         |
| India     | Meghalaya     |
| India     | Rajasthan     |
| Nahi Pata | Kathmandu     |
| Nepal     | Kathmandu     |
| India     | Maharashtra   |
| India     | Karnataka     |
| Nahi Pata | Telangana     |
| India     | Meghalaya     |
| India     | Uttar Pradesh |
| India     | Delhi         |
| India     | Meghalaya     |
| India     | Rajasthan     |
| India     | Telangana     |
| India     | Uttar Pradesh |
| India     | Maharashtra   |

```
In [55]: import numpy as np
```

```
# 1. Convert 'Nahi Pata' back to NaN
dfs['db_customer']['country'] = dfs['db_customer']['country'].replace('Nahi Pata', NaN)

# 2. Create state-to-country mapping dictionary
state_mapping = (
    dfs['db_customer']
    .dropna(subset=['country'])
    .set_index('state')['country']
    .to_dict()
)

# 3. Fill NaN country values using the mapped state
dfs['db_customer']['country'] = dfs['db_customer']['country'].fillna(
    dfs['db_customer']['state'].map(state_mapping)
)
```

```
In [56]: dfs['db_customer'][['country', 'state']]
```

```
Out[56]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>country</th> <th>state</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th>
<th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th>
<th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th>
<th>19</th> <th>20</th> </tbody> <tbody></tbody>
```

|       |               |
|-------|---------------|
| India | Maharashtra   |
| India | Karnataka     |
| India | Delhi         |
| India | Nagaland      |
| India | Delhi         |
| India | Delhi         |
| India | Meghalaya     |
| India | Rajasthan     |
| Nepal | Kathmandu     |
| Nepal | Kathmandu     |
| India | Maharashtra   |
| India | Karnataka     |
| India | Telangana     |
| India | Meghalaya     |
| India | Uttar Pradesh |
| India | Delhi         |
| India | Meghalaya     |
| India | Rajasthan     |
| India | Telangana     |
| India | Uttar Pradesh |
| India | Maharashtra   |

```
In [57]: dfs['db_subscription'].head()
```

```
Out[57]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>customerid</th> <th>subscription_start_date</th>
<th>subscription_type</th> <th>renewal_date</th> <th>plan_type</th>
<th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th>
<th>monthly_charges</th> <th>cltv</th> <th>churn_score</th> </thead> <tbody>
<th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody>
</tbody>
```

|            |            |          |            |          |         |            |                        |       |      |    |
|------------|------------|----------|------------|----------|---------|------------|------------------------|-------|------|----|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual  | NaN        | NaN                    | 13.99 | 627  | 12 |
| 0003-MKNFE | 2020-08-01 | Paid     | 2024-08-01 | Premium  | Annual  | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 |
| 0004-TLHLJ | 2022-11-20 | Organic  | 2025-11-20 | Basic    | Monthly | NaN        | NaN                    | 6.99  | 210  | 34 |
| 0011-IGKFF | 2019-05-10 | Paid     | 2025-05-10 | Premium  | Annual  | NaN        | NaN                    | 22.99 | 1725 | 8  |
| 0013-EXCHZ | 2023-01-05 | Refferal | 2024-01-05 | Standard | Monthly | 2024-02-28 | Too expensive          | 13.99 | 195  | 88 |

```
In [58]: dfs['db_subscription'].info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   customerid                           21 non-null    str
1   subscription_start_date               21 non-null    str
2   subscription_type                     21 non-null    str
3   renewal_date                         21 non-null    str
4   plan_type                            21 non-null    str
5   contract_type                        21 non-null    str
6   cancellation_date                     6 non-null     str
7   cancellation_reason                   6 non-null     str
8   monthly_charges                      21 non-null    float64
9   cltv                                 21 non-null    int64
10  churn_score                           21 non-null    int64
dtypes: float64(1), int64(2), str(8)
memory usage: 1.9 KB
```

```
In [59]: date_col = ['subscription_start_date', 'renewal_date', 'cancellation_date']

dfs['db_subscription'][date_col] = dfs['db_subscription'][date_col].apply(pd.to_datetime)
dfs['db_subscription'].info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   customerid                           21 non-null     str
1   subscription_start_date              21 non-null     datetime64[us]
2   subscription_type                     21 non-null     str
3   renewal_date                         21 non-null     datetime64[us]
4   plan_type                            21 non-null     str
5   contract_type                        21 non-null     str
6   cancellation_date                    6 non-null      datetime64[us]
7   cancellation_reason                   6 non-null      str
8   monthly_charges                      21 non-null     float64
9   cltv                                 21 non-null     int64
10  churn_score                           21 non-null     int64
dtypes: datetime64[us](3), float64(1), int64(2), str(5)
memory usage: 1.9 KB
```

```
In [60]: dfs['db_support'].head()
```

```
Out[60]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>customerid</th> <th>complaint_date</th> <th>escalations</th>
<th>csat_score</th> <th>col_1</th> <th>comment</th> </thead> <tbody>
<th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody>
</tbody>
```

|            |                     |   |    |      |                   |
|------------|---------------------|---|----|------|-------------------|
| 0003-MKNFE | 2024-08-28 00:00:00 | N | 60 | None | service issue     |
| 0003-MKNFE | 2024-08-28 00:00:00 | Y | 10 | None | demaned refund    |
| 0013-EXCHZ | 2024-01-20 00:00:00 | Y | 20 | None | NaN               |
| 0013-MHZWF | 2025-03-18 00:00:00 | N | 90 | None | guidance to renew |
| 0013-SMEOE | 2024-11-01 00:00:00 | N | 30 | None | NaN               |

```
In [61]: dfs['db_support'].info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerid            9 non-null     str
1   complaint_date        9 non-null     str
2   escalations           9 non-null     str
3   csat_score            9 non-null     int64
4   col_1                 0 non-null     object
5   comment               4 non-null     str
dtypes: int64(1), object(1), str(4)
memory usage: 564.0+ bytes
```

```
In [62]: dfs['db_support'].drop(columns=['col_1', 'comment'], inplace=True)
dfs['db_support']['complaint_date'] = pd.to_datetime(dfs['db_support']['complain
dfs['db_support'].info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerid            9 non-null     str
1   complaint_date        9 non-null     datetime64[us]
2   escalations           9 non-null     str
3   csat_score            9 non-null     int64
dtypes: datetime64[us](1), int64(1), str(2)
memory usage: 420.0 bytes
```

In [ ]:

In [63]: *#Feature Engineering and Data Analysis*

In [64]: *# create a new col using existing col - churn flag*  
dfs['db\_subscription']['churn\_flag'] = np.where(dfs['db\_subscription']['cancell

In [65]: dfs['db\_support'].head()

Out[65]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe  
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>  
<th></th> <th>customerid</th> <th>complaint\_date</th> <th>escalations</th>  
<th>csat\_score</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th>  
<th>3</th> <th>4</th> </tbody> <tbody></tbody>

|            |            |   |    |
|------------|------------|---|----|
| 0003-MKNFE | 2024-08-28 | N | 60 |
|------------|------------|---|----|

|            |            |   |    |
|------------|------------|---|----|
| 0003-MKNFE | 2024-08-28 | Y | 10 |
|------------|------------|---|----|

|            |            |   |    |
|------------|------------|---|----|
| 0013-EXCHZ | 2024-01-20 | Y | 20 |
|------------|------------|---|----|

|            |            |   |    |
|------------|------------|---|----|
| 0013-MHZWF | 2025-03-18 | N | 90 |
|------------|------------|---|----|

|            |            |   |    |
|------------|------------|---|----|
| 0013-SMEOE | 2024-11-01 | N | 30 |
|------------|------------|---|----|

In [66]: dfs['db\_subscription'].head()

```
Out[66]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>customerid</th> <th>subscription_start_date</th>
<th>subscription_type</th> <th>renewal_date</th> <th>plan_type</th>
<th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th>
<th>monthly_charges</th> <th>cltv</th> <th>churn_score</th> <th>churn_flag</th>
</thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th>
</tbody> <tbody></tbody>
```

|            |            |          |            |          |         |            |                        |       |      |    |   |
|------------|------------|----------|------------|----------|---------|------------|------------------------|-------|------|----|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual  | NaT        | NaN                    | 13.99 | 627  | 12 | C |
| 0003-MKNFE | 2020-08-01 | Paid     | 2024-08-01 | Premium  | Annual  | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 | 1 |
| 0004-TLHLJ | 2022-11-20 | Organic  | 2025-11-20 | Basic    | Monthly | NaT        | NaN                    | 6.99  | 210  | 34 | C |
| 0011-IGKFF | 2019-05-10 | Paid     | 2025-05-10 | Premium  | Annual  | NaT        | NaN                    | 22.99 | 1725 | 8  | C |
| 0013-EXCHZ | 2023-01-05 | Refferal | 2024-01-05 | Standard | Monthly | 2024-02-28 | Too expensive          | 13.99 | 195  | 88 | 1 |

```
In [67]: dfs['db_subscription'].merge(dfs['db_customer'], on='customerid', how='left')
```

```
Out[67]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>customerid</th> <th>subscription_start_date</th>
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<th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th>
<th>dob</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th>
<th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th>
<th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th>
<th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody>
<tbody></tbody>
```

|            |            |          |            |          |         |            |                        |       |      |    |
|------------|------------|----------|------------|----------|---------|------------|------------------------|-------|------|----|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual  | NaT        | NaN                    | 13.99 | 627  | 12 |
| 0003-MKNFE | 2020-08-01 | Paid     | 2024-08-01 | Premium  | Annual  | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 |
| 0004-TLHLJ | 2022-11-20 | Organic  | 2025-11-20 | Basic    | Monthly | NaT        | NaN                    | 6.99  | 210  | 34 |
| 0011-IGKFF | 2019-05-10 | Paid     | 2025-05-10 | Premium  | Annual  | NaT        | NaN                    | 22.99 | 1725 | 8  |
| 0013-EXCHZ | 2023-01-05 | Refferal | 2024-01-05 | Standard | Monthly | 2024-02-28 | Too expensive          | 13.99 | 195  | 88 |
| 0013-MHZWF | 2022-06-18 | Paid     | 2025-06-18 | Standard | Annual  | NaT        | NaN                    | 17.99 | 720  | 22 |
| 0013-SMEOE | 2021-09-30 | Refferal | 2024-09-30 | Basic    | Monthly | 2024-11-15 | Not enough content     | 8.99  | 230  | 79 |
| 0014-BMAQU | 2020-02-14 | Organic  | 2025-02-14 | Premium  | Annual  | NaT        | NaN                    | 22.99 | 1840 | 5  |
| 0015-UOCOJ | 2023-07-22 | Organic  | 2024-07-22 | Standard | Monthly | NaT        | NaN                    | 13.99 | 240  | 34 |
| 0016-QLJIS | 2022-04-03 | Organic  | 2025-04-03 | Basic    | Annual  | NaT        | NaN                    | 6.99  | 335  | 41 |
| 0017-DINOC | 2021-12-01 | Organic  | 2025-12-01 | Premium  | Annual  | NaT        | NaN                    | 22.99 | 1610 | 14 |
| 0017-IUDMW | 2023-03-17 | Refferal | 2024-03-17 | Standard | Monthly | 2024-05-01 | Poor streaming quality | 7.99  | 270  | 83 |
| 0018-NYROU | 2020-10-09 | Organic  | 2025-10-09 | Standard | Annual  | NaT        | NaN                    | 13.99 | 980  | 19 |
| 0019-EFAEP | 2022-08-25 | Refferal | 2024-08-25 | Basic    | Monthly | 2024-10-31 | Switched to competitor | 12.99 | 160  | 76 |

|            |            |          |            |          |         |            |                        |       |      |    |
|------------|------------|----------|------------|----------|---------|------------|------------------------|-------|------|----|
| 0019-GFNTW | 2019-11-11 | Paid     | 2025-11-11 | Premium  | Annual  | NaT        | NaN                    | 92.99 | 2185 | 3  |
| 0020-INWCK | 2023-05-06 | Organic  | 2025-05-06 | Standard | Monthly | NaT        | NaN                    | 13.99 | 640  | 58 |
| 0020-JDNXP | 2022-01-19 | Organic  | 2024-01-19 | Premium  | Annual  | NaT        | NaN                    | 20.99 | 550  | 62 |
| 0021-IKXGC | 2021-07-07 | Paid     | 2025-07-07 | Standard | Annual  | NaT        | NaN                    | 13.99 | 840  | 27 |
| 0022-TCJCI | 2023-09-14 | Refferal | 2024-09-14 | Basic    | Monthly | 2024-09-14 | Forgot to cancel trial | 16.99 | 42   | 99 |
| 0023-HGHWL | 2020-06-23 | Organic  | 2025-06-23 | Premium  | Annual  | NaT        | NaN                    | 22.99 | 1955 | 7  |
| 0023-UYUPN | 2022-12-31 | Paid     | 2025-12-31 | Standard | Monthly | NaT        | NaN                    | 13.99 | 790  | 47 |

```
In [77]: df = (dfs['db_subscription']
              .merge(dfs['db_customer'], on='customerid', how='left')
              .merge(dfs['db_support'], on='customerid', how='left'))
```

```
In [76]: dfs['db_subscription'].shape
```

```
Out[76]: (21, 12)
```

```
In [78]: df.shape
```

```
Out[78]: (23, 20)
```

```
In [79]: dfs['db_subscription']['customerid'].nunique()
```

```
Out[79]: 21
```

```
In [80]: dfs['db_customer']['customerid'].nunique()
```

```
Out[80]: 21
```

```
In [81]: dfs['db_support']['customerid'].nunique()
```

```
Out[81]: 7
```

```
In [82]: dfs['db_support']['customerid'].size
```

```
Out[82]: 9
```

```
In [83]: dfs['db_support']
```

```
Out[83]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>customerid</th> <th>complaint_date</th> <th>escalations</th>
<th>csat_score</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th>
<th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> </tbody>
<tbody></tbody>
```

```
0003-MKNFE 2024-08-28 N 60
```

```
0003-MKNFE 2024-08-28 Y 10
```

```
0013-EXCHZ 2024-01-20 Y 20
```

```
0013-MHZWF 2025-03-18 N 90
```

```
0013-SMEOE 2024-11-01 N 30
```

```
0017-IUDMW 2024-04-10 Y 25
```

```
0019-EFAEP 2024-09-27 Y 30
```

```
0022-TCJCI 2024-09-13 Y 10
```

```
0022-TCJCI 2024-09-14 N 90
```

```
In [85]: dfs['db_support']['complaint_count']=dfs['db_support'].groupby('customerid')['cu
```

```
In [86]: df.to_csv('exported_churn_data.csv',index=False)
```

```
In [ ]:
```

```
In [88]: #Data Analysis
#1.Churn Rate
```

```
In [91]: df.columns
```

```
Out[91]: Index(['customerid', 'subscription_start_date', 'subscription_type',
               'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
               'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
               'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
               'complaint_date', 'escalations', 'csat_score'],
              dtype='str')
```

```
In [95]: churn_rate = df['churn_flag'].mean()*100
print("Churn rate = ", round(churn_rate,2),"%")
```

```
Churn rate = 34.78 %
```

```
In [96]: #2. Retention Rate
```

```
In [98]: retention_rate = 100- churn_rate
print("retention_rate = ", round(retention_rate,2),"%")
```

```
retention_rate = 65.22 %
```

```
In [99]: df.head(2)
```

```
Out[99]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
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<th>monthly_charges</th> <th>cltv</th> <th>churn_score</th> <th>churn_flag</th>
<th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th>
<th>dob</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th>
</thead> <tbody> <th>0</th> <th>1</th> </tbody> <tbody></tbody>
```

|            |            |          |            |          |        |            |                        |       |      |    |   |
|------------|------------|----------|------------|----------|--------|------------|------------------------|-------|------|----|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual | NaT        | NaN                    | 13.99 | 627  | 12 | 0 |
| 0003-MKNFE | 2020-08-01 | Paid     | 2024-08-01 | Premium  | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 | 1 |



```
In [100... df.groupby('plan_type')['churn_flag'].mean()
```

```
Out[100... plan_type
Basic      0.666667
Premium    0.250000
Standard   0.222222
Name: churn_flag, dtype: float64
```

```
In [101... # 3. Churn by Plan type
churn_by_plan = (df.groupby('plan_type')['churn_flag'].mean().mul(100).round(2).
print(churn_by_plan)
```

```
plan_type  churn_rate_pct
0    Basic             66.67
1    Premium            25.00
2    Standard           22.22
```

```
In [105... arpu = df['monthly_charges'].mean()
print('ARPU =',round(arpu,2))
```

ARPU = 18.51

```
In [106... df.columns
```

```
Out[106... Index(['customerid', 'subscription_start_date', 'subscription_type',
'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
'complaint_date', 'escalations', 'csat_score'],
dtype='str')
```

```
In [107... df.head(2)
```

```
Out[107... <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>customerid</th> <th>subscription_start_date</th>
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<th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th>
<th>monthly_charges</th> <th>cltv</th> <th>churn_score</th> <th>churn_flag</th>
<th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th>
<th>dob</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th>
</thead> <tbody> <th>0</th> <th>1</th> </tbody> <tbody></tbody>
```

|            |            |          |            |          |        |            |                        |       |      |    |   |
|------------|------------|----------|------------|----------|--------|------------|------------------------|-------|------|----|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual | NaT        | NaN                    | 13.99 | 627  | 12 | 0 |
| 0003-MKNFE | 2020-08-01 | Paid     | 2024-08-01 | Premium  | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 | 1 |

```
In [111... #6. Avg customer tenure
# count of days user has used our service: cancellation date else current date
today = pd.Timestamp.today()

df['tenure_days'] = np.where(
    df['cancellation_date'].notna(),

    (df['cancellation_date'] - df['subscription_start_date']).dt.days,

    (today - df['subscription_start_date']).dt.days
)

avg_tenure = df['tenure_days'].mean()
print("Avg Tenure (Days) = ", round(avg_tenure), 0)
```

Avg Tenure (Days) = 1488 0

```
In [114... # Revenue at risk - revenue lost from churned users
revenue_at_risk = df.loc[df['churn_flag']==1, 'monthly_charges'].sum()
print("Revenue at risk (Rs 'k')=", revenue_at_risk)
```

Revenue at risk (Rs 'k')= 103.91999999999999

```
In [115... # 8. Esclation Rate
escalation_rate = (df['escalations']=='Y').mean()*100
print("Esclation Rate = ", round(escalation_rate, 2), "%")
```

Esclation Rate = 21.74 %

```
In [120... print(df.columns.tolist())
```

```
['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'p
lan_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly
charges', 'cltv', 'churn_score', 'churn_flag', 'customer_name', 'country', 'stat
e', 'gender', 'dob', 'complaint_date', 'escalations', 'csat_score', 'tenure_day
s']
```

```
In [121... df['complaint_count'] = df['complaint_date'].notna().astype(int)
```

```
In [122... # Check value counts to verify the new column
print(df['complaint_count'].value_counts())
df.head()
```

```
complaint_count
```

```
0    14
```

```
1     9
```

```
Name: count, dtype: int64
```

```
Out[122... <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
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<th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th>
<th>monthly_charges</th> <th>cltv</th> <th>...</th> <th>customer_name</th>
<th>country</th> <th>state</th> <th>gender</th> <th>dob</th>
<th>complaint_date</th> <th>escalations</th> <th>csat_score</th>
<th>tenure_days</th> <th>complaint_count</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>
```

|            |            |          |            |          |         |            |                        |       |      |     |    |
|------------|------------|----------|------------|----------|---------|------------|------------------------|-------|------|-----|----|
| 0002-ORFBO | 2021-03-15 | Referral | 2025-03-15 | Standard | Annual  | NaT        | NaN                    | 13.99 | 627  | ... | ke |
| 0003-MKNFE | 2020-08-01 | Paid     | 2024-08-01 | Premium  | Annual  | 2024-09-10 | Switched to competitor | 12.99 | 1150 | ... | ra |
| 0003-MKNFE | 2020-08-01 | Paid     | 2024-08-01 | Premium  | Annual  | 2024-09-10 | Switched to competitor | 12.99 | 1150 | ... | ra |
| 0004-TLHLJ | 2022-11-20 | Organic  | 2025-11-20 | Basic    | Monthly | NaT        | NaN                    | 6.99  | 210  | ... |    |
| 0011-IGKFF | 2019-05-10 | Paid     | 2025-05-10 | Premium  | Annual  | NaT        | NaN                    | 22.99 | 1725 | ... | m  |

<p>5 rows × 22 columns</p>



```
In [128... # 9. Avg Complaint Per User
avg_complaints = df['complaint_count'].sum() / df['customerid'].nunique()
print("Avg complaint per user =",round(avg_complaints,2))
```

```
Avg complaint per user = 0.43
```

```
In [126... df.columns
```

```
Out[126... Index(['customerid', 'subscription_start_date', 'subscription_type',
      'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
      'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
      'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
      'complaint_date', 'escalations', 'csat_score', 'tenure_days',
      'complaint_count'],
      dtype='str')
```

```
In [139... # 10. Correlation Escalation vs Churn
# Convert 'Y'/'N' to binary numeric values
df['escalation_flag'] = np.where(df['escalations'] == 'Y', 1, 0)
```

```
# Create clean DataFrame and calculate correlation
corr_df = df[['escalation_flag', 'churn_flag']].dropna()
correlation = corr_df['escalation_flag'].corr(corr_df['churn_flag'])

print("Correlation Escalation vs Churn =", round(correlation, 2))
```

Correlation Escalation vs Churn = 0.72

In [140...] `df['churn_score'].head()`

Out[140...] 

|   |    |
|---|----|
| 0 | 12 |
| 1 | 91 |
| 2 | 91 |
| 3 | 34 |
| 4 | 8  |

  
Name: churn\_score, dtype: int64

In [143...] 

```
#Churn risk - create a cloumn using existing col
conditions = [
    (df['churn_score'] < 50),
    (df['churn_score'] >= 50) & (df['churn_score'] < 70),
    (df['churn_score'] >= 70)
]
choices = ['low', 'med', 'high']
df['churn_risk'] = np.select(conditions, choices, default='unknown')
```

In [145...] `df[['churn_risk', 'churn_score']].tail()`

Out[145...] 

|    | churn_risk | churn_score |
|----|------------|-------------|
| 18 | low        | 27          |
| 19 | high       | 99          |
| 20 | high       | 99          |
| 21 | low        | 7           |
| 22 | low        | 47          |

In [146...] 

```
#Visualization using matplotlib
```

In [149...] `df_visual = df.copy()`

In [150...] `df_visual.head()`

```
Out[150...] <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
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<th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th>
<th>monthly_charges</th> <th>cltv</th> <th>...</th> <th>state</th>
<th>gender</th> <th>dob</th> <th>complaint_date</th> <th>escalations</th>
<th>csat_score</th> <th>tenure_days</th> <th>complaint_count</th>
<th>escalation_flag</th> <th>churn_risk</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>
```

|            |            |          |            |          |         |            |                        |       |      |     |   |
|------------|------------|----------|------------|----------|---------|------------|------------------------|-------|------|-----|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual  | NaT        | NaN                    | 13.99 | 627  | ... | M |
| 0003-MKNFE | 2020-08-01 | Paid     | 2024-08-01 | Premium  | Annual  | 2024-09-10 | Switched to competitor | 12.99 | 1150 | ... |   |
| 0003-MKNFE | 2020-08-01 | Paid     | 2024-08-01 | Premium  | Annual  | 2024-09-10 | Switched to competitor | 12.99 | 1150 | ... |   |
| 0004-TLHLJ | 2022-11-20 | Organic  | 2025-11-20 | Basic    | Monthly | NaT        | NaN                    | 6.99  | 210  | ... |   |
| 0011-IGKFF | 2019-05-10 | Paid     | 2025-05-10 | Premium  | Annual  | NaT        | NaN                    | 22.99 | 1725 | ... |   |

<p>5 rows × 24 columns</p>

```
In [151...] df.columns
```

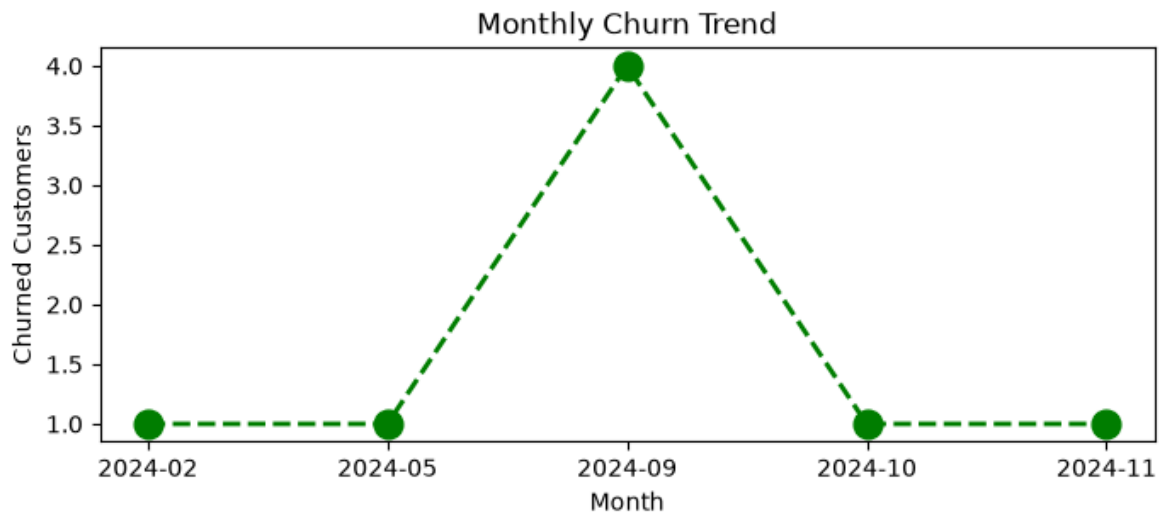
```
Out[151...] Index(['customerid', 'subscription_start_date', 'subscription_type',
      'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
      'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
      'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
      'complaint_date', 'escalations', 'csat_score', 'tenure_days',
      'complaint_count', 'escalation_flag', 'churn_risk'],
      dtype='str')
```

```
In [163...] import matplotlib.pyplot as plt

# Monthly Churn Trend(Time Series KPI)
df_visual['cancellation_month'] = df_visual['cancellation_date'].dt.to_period('M')

churn_trend = df_visual[df_visual['churn_flag'] == 1].groupby('cancellation_month')

plt.figure(figsize=(8, 3))
plt.plot(churn_trend.index.astype(str), churn_trend.values, color='green', marker='o')
plt.title('Monthly Churn Trend')
plt.xlabel('Month')
plt.ylabel('Churned Customers')
plt.show()
```



In [156...

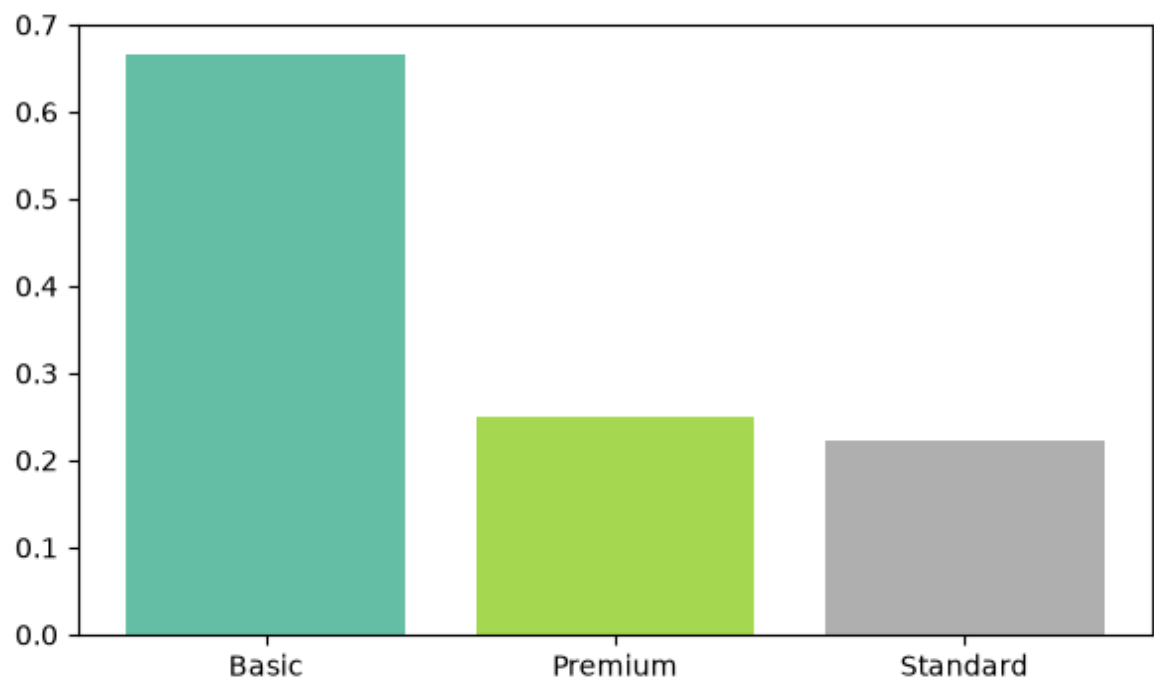
Out[156... cancellation\_month

|         |   |
|---------|---|
| 2024-02 | 1 |
| 2024-05 | 1 |
| 2024-09 | 4 |
| 2024-10 | 1 |
| 2024-11 | 1 |

Freq: M, dtype: int64

In [168...

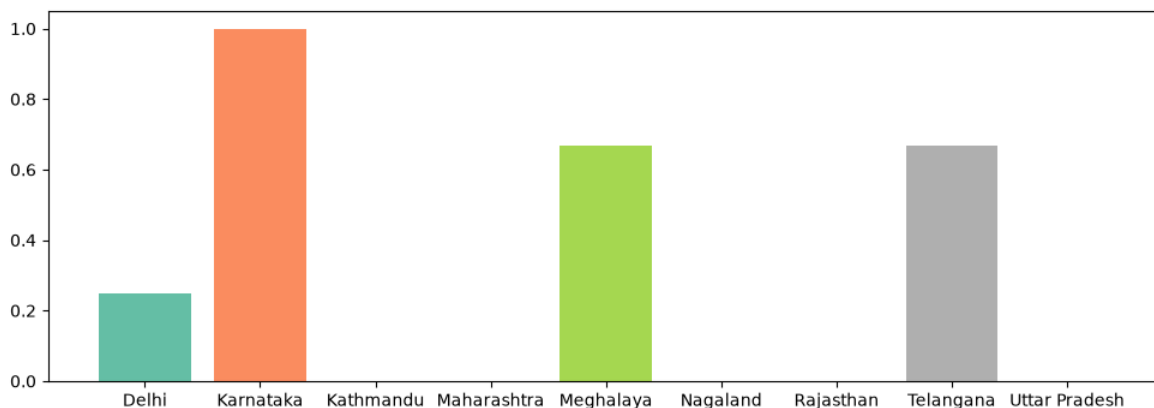
```
# 4.2 Churn by Plan type
churn_plan = df_visual.groupby('plan_type')['churn_flag'].mean()
# colors=['yellow', 'purple', 'blue']
colors= plt.cm.Set2(np.linspace(0,1,len(churn_plan)))
plt.figure(figsize=(7, 4))
plt.bar(churn_plan.index, churn_plan.values,color=colors)
plt.show()
```



In [170...

```
# 4.2 Churn by States
churn_plan = df_visual.groupby('state')['churn_flag'].mean()
# colors=['yellow', 'purple', 'blue']
colors= plt.cm.Set2(np.linspace(0,1,len(churn_plan)))
```

```
plt.figure(figsize=(12, 4))
plt.bar(churn_plan.index, churn_plan.values,color=colors)
plt.show()
```



In [173...] *# Visualization using seaborn*

In [174...] *#encoding convert string to numeric values so that we can find corealtion between df\_visual.columns*

Out[174...] Index(['customerid', 'subscription\_start\_date', 'subscription\_type', 'renewal\_date', 'plan\_type', 'contract\_type', 'cancellation\_date', 'cancellation\_reason', 'monthly\_charges', 'cltv', 'churn\_score', 'churn\_flag', 'customer\_name', 'country', 'state', 'gender', 'dob', 'complaint\_date', 'escalations', 'csat\_score', 'tenure\_days', 'complaint\_count', 'escalation\_flag', 'churn\_risk', 'cancellation\_month'], dtype='str')

In [176...] *# Corrected syntax (added missing quote after 'churn\_score')*  
df\_encoded = df\_visual[['plan\_type', 'contract\_type', 'churn\_score', 'churn\_flag', 'churn\_risk', 'escalations', 'csat\_score', 'tenure\_days', 'complaint\_count', 'escalation\_flag', 'cancellation\_month']]

Out[176...] <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>  
<th></th> <th>plan\_type</th> <th>contract\_type</th> <th>churn\_score</th>  
<th>churn\_flag</th> <th>churn\_risk</th> <th>escalations</th> </thead> <tbody>  
<th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody>  
</tbody>

|          |         |    |   |      |     |
|----------|---------|----|---|------|-----|
| Standard | Annual  | 12 | 0 | low  | NaN |
| Premium  | Annual  | 91 | 1 | high | N   |
| Premium  | Annual  | 91 | 1 | high | Y   |
| Basic    | Monthly | 34 | 0 | low  | NaN |
| Premium  | Annual  | 8  | 0 | low  | NaN |

In [177...] *# 1. Select full column subset*  
df\_encoded = df\_visual[['plan\_type', 'contract\_type', 'churn\_score', 'churn\_flag', 'churn\_risk', 'escalations', 'csat\_score', 'tenure\_days', 'complaint\_count', 'escalation\_flag', 'cancellation\_month']]

*# 2. Encode categorical string columns to numeric integers*  
categorical\_cols = ['plan\_type', 'contract\_type', 'churn\_risk', 'escalations']  
for col in categorical\_cols:

```
df_encoded[col] = df_encoded[col].astype('category').cat.codes

# 3. Preview top 5 rows
df_encoded.head()
```

Out[177... <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>plan\_type</th> <th>contract\_type</th> <th>churn\_score</th> <th>churn\_flag</th> <th>churn\_risk</th> <th>escalations</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody>

```
2  0  12  0  1  -1
```

```
1  0  91  1  0  0
```

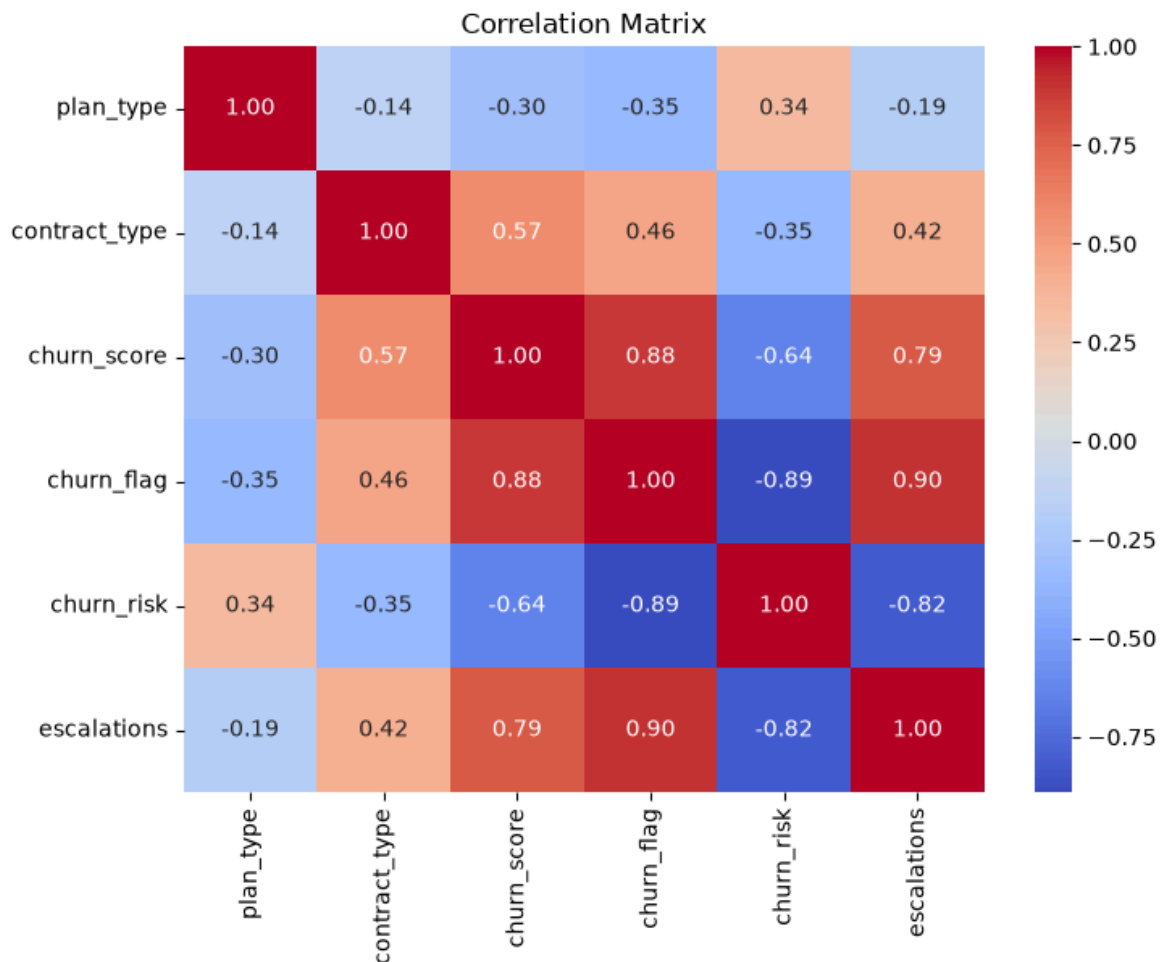
```
1  0  91  1  0  1
```

```
0  1  34  0  1  -1
```

```
1  0   8  0  1  -1
```

In [182... **import** matplotlib.pyplot **as** plt  
**import** seaborn **as** sns

```
# Heatmap Correlation matrix
plt.figure(figsize=(8, 6))
sns.heatmap(df_encoded.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Matrix')
plt.show()
```



```
In [183...] df_visual['churn_risk'].unique()
```

```
Out[183...] <StringArray>
['low', 'high', 'med']
Length: 3, dtype: str
```

```
In [185...] import pandas as pd

# 1. Select subset with .copy() to avoid SettingWithCopyWarning
df_encoded = df_visual[['plan_type', 'contract_type', 'churn_score', 'churn_flag', 'churn_risk', 'escalations']].copy()

# 2. Define custom priority orders
order_mappings = {
    'plan_type': ['Basic', 'Standard', 'Premium'],
    'contract_type': ['Monthly', 'Annual'],
    'churn_risk': ['low', 'med', 'high'] # Ensure case matches your data ('Low' vs 'low')
}

# 3. Apply ordinal encoding
for col, order in order_mappings.items():
    df_encoded[col] = pd.Categorical(df_encoded[col], categories=order, ordered=True)

# 4. Map 'escalations' to numeric (1/0) so it works in correlation matrix
df_encoded['escalations'] = (df_encoded['escalations'] == 'Y').astype(int)
```

```
In [187...] df_visual[['plan_type', 'contract_type', 'churn_score', 'churn_flag', 'churn_risk', 'escalations']]
```

Out[187... <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>plan\_type</th> <th>contract\_type</th> <th>churn\_score</th> <th>churn\_flag</th> <th>churn\_risk</th> <th>escalations</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody> </tbody>

|          |        |    |   |     |     |
|----------|--------|----|---|-----|-----|
| Standard | Annual | 12 | 0 | low | NaN |
|----------|--------|----|---|-----|-----|

|         |        |    |   |      |   |
|---------|--------|----|---|------|---|
| Premium | Annual | 91 | 1 | high | N |
|---------|--------|----|---|------|---|

|         |        |    |   |      |   |
|---------|--------|----|---|------|---|
| Premium | Annual | 91 | 1 | high | Y |
|---------|--------|----|---|------|---|

|       |         |    |   |     |     |
|-------|---------|----|---|-----|-----|
| Basic | Monthly | 34 | 0 | low | NaN |
|-------|---------|----|---|-----|-----|

|         |        |   |   |     |     |
|---------|--------|---|---|-----|-----|
| Premium | Annual | 8 | 0 | low | NaN |
|---------|--------|---|---|-----|-----|

In [186... df\_encoded.head()

Out[186... <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>plan\_type</th> <th>contract\_type</th> <th>churn\_score</th> <th>churn\_flag</th> <th>churn\_risk</th> <th>escalations</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody> </tbody>

|   |   |    |   |   |   |
|---|---|----|---|---|---|
| 1 | 1 | 12 | 0 | 0 | 0 |
|---|---|----|---|---|---|

|   |   |    |   |   |   |
|---|---|----|---|---|---|
| 2 | 1 | 91 | 1 | 2 | 0 |
|---|---|----|---|---|---|

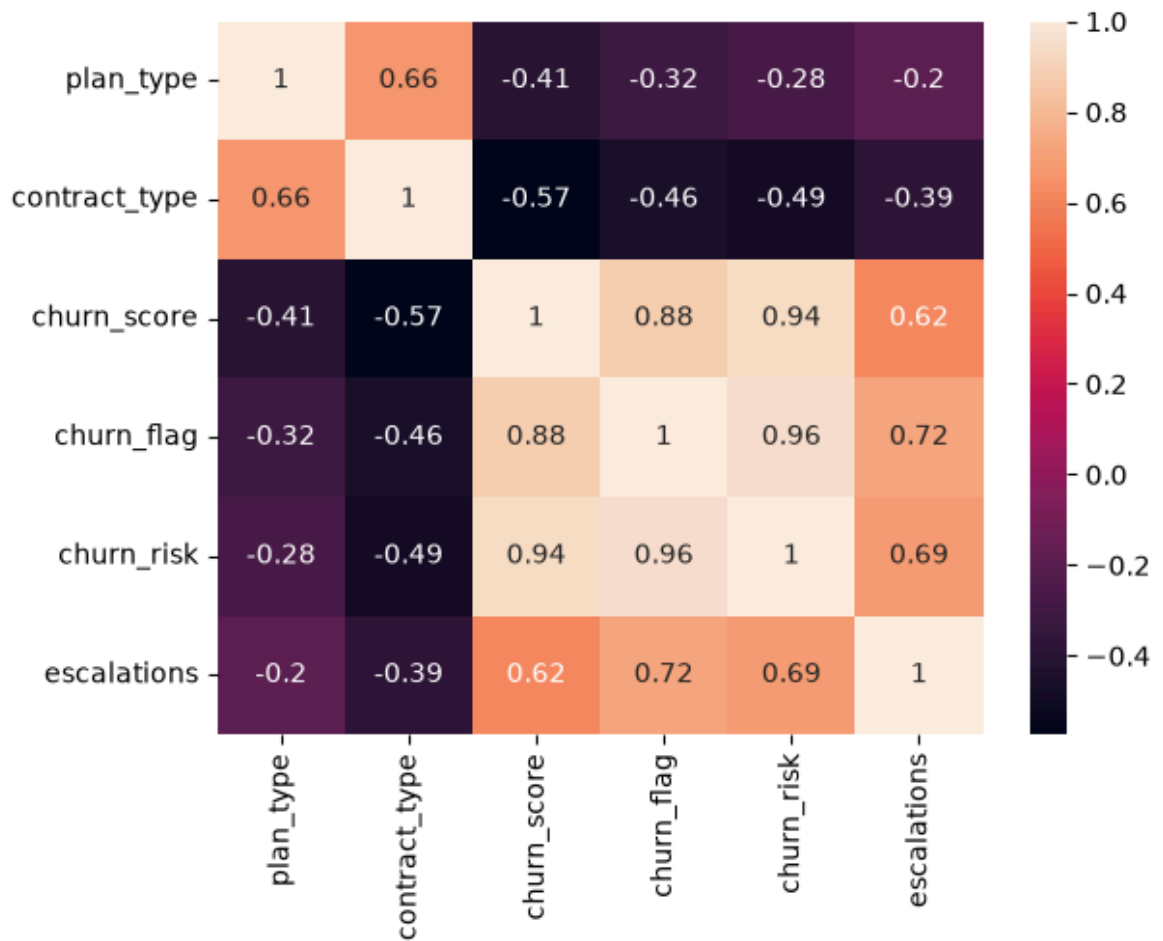
|   |   |    |   |   |   |
|---|---|----|---|---|---|
| 2 | 1 | 91 | 1 | 2 | 1 |
|---|---|----|---|---|---|

|   |   |    |   |   |   |
|---|---|----|---|---|---|
| 0 | 0 | 34 | 0 | 0 | 0 |
|---|---|----|---|---|---|

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 2 | 1 | 8 | 0 | 0 | 0 |
|---|---|---|---|---|---|

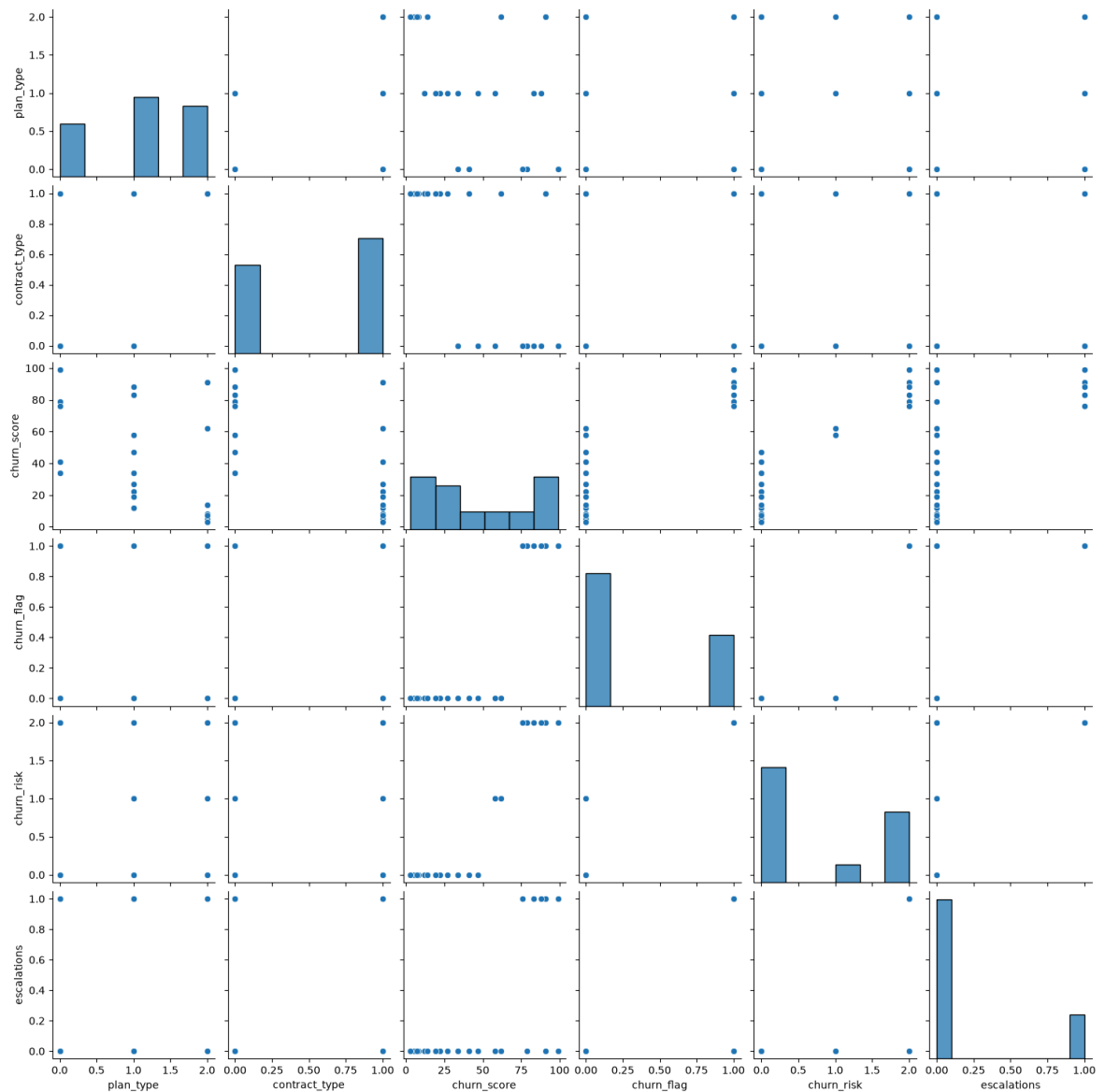
In [188... *#Heatmap Corelation matrix*  
sns.heatmap(df\_encoded.corr(), annot=True)

Out[188... <Axes: >



In [190...

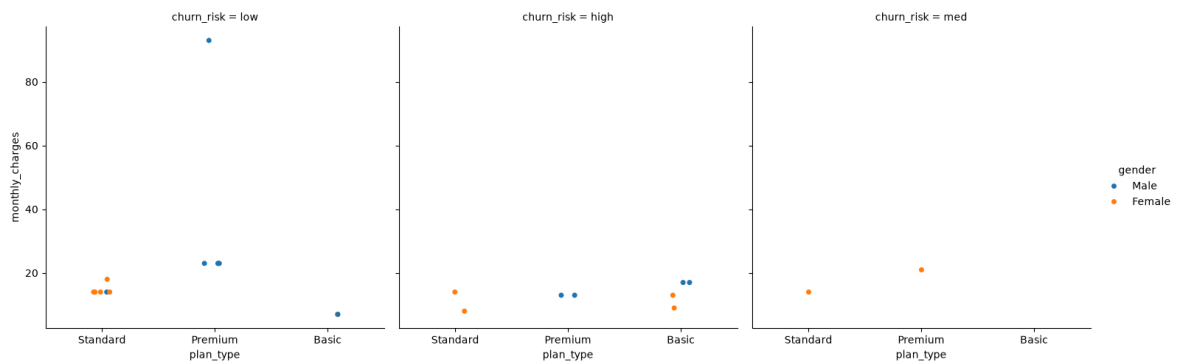
```
# pairplot  
sns.pairplot(df_encoded)  
plt.show()
```



In [191... `#catplot/FacegridPlot -multi-dim comparison`

```
sns.catplot(data=df_visual,
             x='plan_type',
             y='monthly_charges',
             hue='gender',
             col='churn_risk')
```

Out[191... `<seaborn.axisgrid.FacetGrid at 0x1d13e8f1e80>`



In [195... `pivot_res = pd.pivot_table(
df_visual,
index='plan_type',`

```

values=['monthly_charges', 'customerid', 'churn_flag'],
aggfunc={
    'monthly_charges': 'sum',
    'customerid': 'nunique',
    'churn_flag': 'mean'
}
)

pivot_res

```

```

Out[195... <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe
tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>churn_flag</th> <th>customerid</th> <th>monthly_charges</th>
<th>plan_type</th> <th></th> <th></th> <th></th> </thead> <tbody>
<th>Basic</th> <th>Premium</th> <th>Standard</th> </tbody> <tbody></tbody>

0.666667    5    69.94
0.250000    7   231.92
0.222222    9   123.91

```

```
In [ ]: # working with sql in python(pandas)
```

```

In [196... #create db in sql
conn = sqlite3.connect('test_database.sqlite')

#table details
conn.execute("CREATE TABLE users (first_name TEXT, country TEXT, budget INTEGER )"

#commit and save
conn.commit()

```

```

In [199... # insert data
cursor = conn.cursor()

# write sql query to insert records in sql table
cursor.execute(
    """
    INSERT INTO users VALUES
        ('Madhav', 'India', 5000),
        ('Rishav', 'Germany', 2500),
        ('Vishaka', 'India', 3500)
    """
)
# commit and save
conn.commit()

print("Data Inserted Successfully")

```

Data Inserted Successfully

```

In [202... import sqlite3
import pandas as pd

# Connect to database file
conn = sqlite3.connect('test_database_sqlite.db')
cursor = conn.cursor()

```

```

# 1. Create table schema if it doesn't exist
cursor.execute('''
    CREATE TABLE IF NOT EXISTS users (
        name TEXT,
        country TEXT,
        balance REAL
    )
''')

# 2. Insert records cleanly
cursor.execute('''
    INSERT INTO users (name, country, balance) VALUES
        ('Madhav', 'India', 5000),
        ('Rishav', 'Germany', 2500),
        ('Vishaka', 'India', 3500)
''')
conn.commit()

# 3. Read back into Pandas DataFrame
query = "SELECT * FROM users"
df_results = pd.read_sql(query, conn)
conn.close()

df_results.head()

```

Out[202... <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>name</th> <th>country</th> <th>balance</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> </tbody> <tbody></tbody>

|         |         |        |
|---------|---------|--------|
| Madhav  | India   | 5000.0 |
| Rishav  | Germany | 2500.0 |
| Vishaka | India   | 3500.0 |

In [204...

```

import sqlite3
import pandas as pd

# Re-open connection
conn = sqlite3.connect('test_database_sqlite.db')

# Update column name to 'balance' if that was used in your CREATE/INSERT stateme
query = """
    SELECT country, SUM(balance) AS total_budget
    FROM users
    GROUP BY country
    """

df_agg = pd.read_sql(query, conn)
conn.close()

df_agg

```

Out[204... `<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>country</th> <th>total_budget</th> </thead> <tbody> <th>0</th> <th>1</th> </tbody> <tbody> </tbody>`

|         |        |
|---------|--------|
| Germany | 2500.0 |
| India   | 8500.0 |

In [205... `conn.close()`

In [ ]: